

Eq. (6.4.4), we have

$$\begin{aligned} & \left[\frac{\delta S[\epsilon]}{\delta \epsilon_{a_1}(x_1) \cdots \delta \epsilon_{a_r}(x_r)} \right]_{\epsilon=0} \\ &= (-i)^r \left(\Omega(\infty) \Phi_\beta, O_{a_1}(x_1) \cdots O_{a_r}(x_r) \Omega(-\infty) \Phi_\alpha \right). \end{aligned} \quad (6.4.12)$$

In deriving this result we supposed that $x_1^0 \geq x_2^0 \geq \cdots \geq x_r^0$, so we could just as well replace the product of operators on the right-hand-side with the time-ordered product:

$$\begin{aligned} & \left[\frac{\delta S[\epsilon]}{\delta \epsilon_{a_1}(x_1) \cdots \delta \epsilon_{a_r}(x_r)} \right]_{\epsilon=0} \\ &= (-i)^r \left(\Omega(\infty) \Phi_\beta, T \left\{ O_{a_1}(x_1) \cdots O_{a_r}(x_r) \right\} \Omega(-\infty) \Phi_\alpha \right). \end{aligned} \quad (6.4.13)$$

But now both sides are entirely symmetric (or for fermions antisymmetric) in the a s and x s, so this relation holds whatever the order of the times $x_1^0 \dots x_r^0$. Also, we saw in Section 3.1 that (in the sense of Eq. (3.1.12))

$$\Psi_\beta^\pm = \Omega(\mp\infty) \Phi_\beta. \quad (6.4.14)$$

Hence Eq. (6.4.13) is the desired result (6.4.3).

Problems

1. Consider the theory of a real scalar field ϕ , with interaction (in the interaction picture) $V = g \int d^3x \phi(x)^3/3!$. Calculate the connected S -matrix element for scalar-scalar scattering to second order in g , doing all integrals. Use the results to calculate the differential cross-section for scalar-scalar scattering in the center-of-mass system.
2. Consider a theory involving a neutral scalar field $\phi(x)$ for a boson B and a complex Dirac field $\psi(x)$ for a fermion F , with interaction (in the interaction picture) $V = ig \int d^3x \bar{\psi}(x) \gamma_5 \psi(x) \phi(x)$. Draw all the connected order- g^2 Feynman diagrams and calculate the corresponding S -matrix elements for the processes $F^c + B \rightarrow F^c + B$, $F + F^c \rightarrow F + F^c$, and $F^c + F \rightarrow B + B$ (where F^c is the antiparticle of F). Do all integrals.
3. Consider the theory of a real scalar field $\phi(x)$, with interaction $V = g \int d^3x \phi(x)^4/4!$. Calculate the S -matrix for scalar-scalar scattering to order g , and use the result to calculate the differential scattering cross-section. Calculate the correction terms in the S -matrix for scalar-scalar scattering to order g^2 , expressing the result as an integral over a single four-momentum, but do all x -integrals.